Seldon Manifolds — Decision Making

Operative Continuation, Prospective Deliberation, and Manifold Revision

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DRAFT — draft-2026-09-06.1r11 (6 September 2026). Circulated for comment, not as a finished result. This is the decision-making companion to *Seldon Manifolds — Theory* (referred to below as *Seldon*) and inherits its definitions, together with those of *The Law of Graphic Equalisation — Theory*, the momentum paper, and *Narrativium*. This paper proposes an ontological interpretation of realised decision as embedded continuation, a finite specification of minimum-cost dynamics, and an empirical hypothesis about deliberation as restructuring of operative future geometry. These have distinct burdens of support. The finite construction supplies an implementable instance; it does not establish that human, organisational, or artificial agents universally instantiate its cost rule. Operational commitments are intended points of attack.

Abstract

At a declared agent and action scale, a realised decision is the operative system's action-bearing continuation through its reachable future geometry. Deliberation is a present process concerning unrealised continuations that can change the conditions of subsequent action. This identity places deciding within the system's organisation. The narrower deliberative use of "decision" attributes an action or commitment to cognitive manifold deformation; that causal attribution is inferred, rather than another event added to the realised continuation. It applies the series' principle that provenance is inferred from present traces rather than inherently labelled in an output. A bare set of reachable histories supplies no successor rule; an operative description must also include admissibility, cost fields, standing structure, capacity, and dynamics.

The proposed minimum-continuation hypothesis identifies action with the lowest effective-cost continuation in the operative comparison domain. A finite specification makes that hypothesis implementable while separating local edge cost, state potential, accumulated expenditure, and Narrativium. Local minimisation does not establish a globally cheapest future. Costs and dynamics must be grounded independently of observed choices.

Biological evolution supplies the central structural example: variation, inheritance, and differential reproduction change the organisation that generates subsequent continuations without requiring prospective modelling. Autonomous processing is the installed response organisation operating; hypothetical processing is embedded modelling that can restructure it. This connects to Type 1/Type 2 cognition without equating all continuation with Type 1 or all revision with Type 2. The represented decision grammar retains Narrative, surprise, paid meta-analysis, acceptable target shapes, and present-state authorisation of self-change. Worked finite examples separate representation from operative revision and actual consequences. The account is agent-neutral and proposes independently testable groundings; it claims neither empirical validation nor a universal energy-minimisation theorem.

1 Scope: decision as operative continuation

Seldon defines $\mathcal{S}_{h,\rho}(\mathcal{F})$, the structured representation at horizon h and resolution ρ of future histories reachable under represented admissible interventions and boundary conditions (Definition 6.1). It distinguishes a local admissibility condition from non-zero-horizon future geometry (Proposition 6.2), and represents goals, where held, as acceptable subsets (Definition 11.1). None of those constructions selects an action. This paper adds an account of the operative organisation whose continuation realises action, and of present processes that change its subsequent future geometry.

Definition 1.1 (Realised decision)**.** *For a declared agent boundary, action scale, and causal resolution, a realised decision is the action-bearing transition or history of that embedded system. A represented selection or commitment is a present state that can condition such a transition; its intended action may remain unrealised. The eventual outcome is the consequence of the realised history under its boundary conditions. These uses of decision, commitment, and outcome are kept distinct.*

Remark 1.2 (Identity and scope)*.* The proposed identity is that the realised decision *is* this continuation. The action-bearing organisation, including any selection machinery, belongs to the system. The definition does not imply that every physical transition is a decision; the instance must declare an agent and action scale. It requires neither conscious endorsement nor an observer to make the transition occur. The identity alone does not predict which action occurs, or establish a minimum-cost law.

Definition 1.3 (Deliberative decision attribution)**.** *In the narrower deliberative sense, calling an action or commitment a decision attributes causal influence to a deliberate cognitive process that deformed operative future geometry. The process can change a represented landscape, comparison field, constraint, or installed response organisation. The attribution is a model of that causal history, inferred from available evidence. It is distinct from the action-bearing continuation and from the cognitive process, both of which occur whether or not the attribution is made.*

This distinguishes the paper's broad structural use of realised decision from the familiar deliberative use of the word. In the latter, the word identifies how a continuation was influenced. It does not introduce a separate act of choosing after the cognitive deformation. A decision about a future action can also name the resulting present commitment; the intended action can remain unrealised.

The same visible action can arise from an installed autonomous response or from a cognitively revised organisation. Action alone therefore does not identify deliberative provenance. Behavioural history, process measurements, and self-report can constrain the attribution; a self-model is not guaranteed complete access to its own causes. Evidence of prior thought alone also does not establish that it influenced the action. The causal question concerns operative influence, without requiring that removing the deliberation must always change the outcome: other sufficient causes can yield the same action.

Remark 1.4 (Decision provenance is inferred, not inherent)*.* The deliberative attribution is an application of the series' provenance principle. A response carries no self-authenticating certificate, independent of its supporting relations, identifying it as deliberate, habitual, evolved, trained, or mechanically conditioned. Explicit annotations or authenticated records can nevertheless constrain local attribution under declared recording and trust assumptions. Those provenance terms describe causal relations inferred from the structure and evidence available to a modelling frame. The actual causal relations do not depend on that inference; the inference can be accurate, incomplete, or mistaken. This inherits Graphic Equalisation's foundational definition of relational provenance (Definition 2.2) and its evidence-relative recovery proposition (Proposition 2.4). The

operator-supply result of Proposition 14.3 is another application, rather than the source of this general principle.

The underlying idea is established source-monitoring and identification work, not a new discovery by this paper. In human source monitoring, origins are attributed through memory characteristics and judgment processes, which can operate automatically or deliberately (Johnson et al., 1993). Graphic Equalisation provides a derivation conditional on its relational commitments and the declared evidence model; the present paper applies it to the causal attribution of deliberation.

Present traces and records can strongly constrain a local causal attribution. Their evidential value does not turn provenance into an additional property that an output exposes independently of a model. A verbal explanation is also present structure whose relation to an earlier action must be inferred. The same response can arise through different processes, and the agent's self-model is one modelling frame among others. Self-report therefore does not close the provenance question merely by coming from the actor.

The argument therefore has three levels. The ontological interpretation identifies realised decision with embedded continuation. The formal specification supplies explicit state, fields, and dynamics for a finite instance. The empirical hypothesis asks whether independently grounded instances explain persistence and revision in actual agents. A correct finite construction establishes the second level under its premises; human application needs additional evidence. The paper supplies no neural mechanism or complete account of motor execution. Inference, carrying a policy, and realising an outcome remain different processes within the same system.

2 The decision frame

Definition 2.1 (Decision frame)**.** *For a current frame $\mathcal{F}$, let its decision description be*

$$D(\mathcal{F}) = (A_{\mathcal{F}}, L_{\mathcal{F}}, S_{\mathcal{F}}, \Gamma_{\mathcal{F}}, N_{\mathcal{F}}, V_{\mathcal{F}}, R_{\mathcal{F}}),$$

where $A_{\mathcal{F}}$ is aperture, $L_{\mathcal{F}}$ the represented landscape, $S_{\mathcal{F}}$ the standing Narrative, $\Gamma_{\mathcal{F}}$ represented admissibility, $N_{\mathcal{F}}$ Narrativium, $V_{\mathcal{F}}$ active comparison descriptions, and $R_{\mathcal{F}}$ represented residual.

Aperture and residual are inherited from Graphic Equalisation (Definitions 7.2 and 8.1). In particular, an aperture omission is absent to the frame, rather than automatically a residual. Narrativium is inherited as frame-relative capacity for consequential deformation (Definition 3.1); it is a state measurement and feasibility relation, not expenditure, energy, a conserved balance, or a universal utility.

Definition 2.2 (Represented decision domain)**.** *The domain of represented decision evaluation is $L_{\mathcal{F}}$ only. A candidate is not silently assessed as a negative option when it is outside $A_{\mathcal{F}}$ or absent from $L_{\mathcal{F}}$.*

This restriction concerns in-frame evaluation. Omitted consequences can still occur, and unrepresented influences can still change actual continuation.

Definition 2.3 (Operative description)**.** *At a declared boundary and resolution, an operative description contains state X, a counterfactual transition relation $\mathcal{R}$, admissibility, standing structure, capacity, comparison fields, and a continuation rule b. Its enriched future geometry is the reachable family together with these operative conditions. A bare reachable set is one projection of it. The agent's represented description $D(\mathcal{F})$, an observer's measurement of the operative system, and the actual system are distinct.*

A supplied deterministic grounding can be written

$$X_{t+1} = T(X_t, B_t), \qquad D(\mathcal{F}_t) = \Pi_{\mathcal{F}_t}(X_t, B_t),$$

where X_t, B_t are sufficient for that instance and B_t states relevant containing conditions. The index orders local transitions; it introduces no global clock. The projection is not an additional chooser. If the measured state is insufficient, the observer must use a declared transition relation or an epistemic probability model, rather than report a known unique successor. Such uncertainty does not add stochasticity to the inherited Graph Law.

Remark 2.4 (Complete fixing and nontrivial comparison). When causally complete antecedents, interventions, and containing conditions are fixed, the inherited deterministic construction has one continuation (*Seldon*, Corollary 4.2). A minimum over that singleton has no explanatory force. Nontrivial comparisons concern the family indexed by counterfactual interventions or a partial description that does not fix all determining organisation. Actual mechanisms generating the intervention belong to the operative system. A cost rule that explains their continuation is an additional hypothesis about that family, not a consequence of determinism.

Changing a comparison field while keeping the available actions fixed changes the enriched operative geometry, but not the bare reachable set. Changing the set through a constraint is a different deformation. The claim that changing decision changes the manifold uses this enriched meaning. An observer's redescription without causal interaction need not change any operative conditions.

Remark 2.5 (Agent neutrality and coupling). The definition applies to an individual, business, state, institution, or coalition. At collective scale, aperture may be distributed across offices and information systems; Narrative may be carried by institutional memory, procedures, and commitments; admissibility may be carried by authority and rules; and meta-analysis may require coordination. These are differences in representation and cost, not in decision kind.

A collective's partial frames differ by degree and structure of coupling, not by ontology. Shared representation, communication, authority, binding procedure, memory, commitment, and coordinated capacity can make the partial frames more mutually constraining. A so-called group mind is the high-coupling end of that continuum, not a distinct kind of agent. Where coupling is weak, the proper object of analysis is an interaction among partial frames; where it is strong, an effective collective frame can be used at the operative scale.

3 Biological evolution: changing the organisation that continues

Biological evolution gives a direct example of operative restructuring without prospective deliberation. Heritable variation changes the structures organisms carry; differential reproduction changes which structures are represented in later populations. The resulting organisms realise continuations through those structures. Their response organisation can embody the consequences of earlier selection without representing that selection or reasoning about the future it makes reachable.

At an organism's action scale, inherited and developed organisation conditions responses. At a population scale, evolutionary change alters the distribution of such organisation across generations. These are different boundaries and causal resolutions. Calling an organism's action a realised decision does not require treating a population as a deliberating agent, or calling each mutation a decision.

For a supplied sufficient organism description, write

$$X_{t+1} = T_\theta(X_t, B_t),$$

where θ denotes inherited components of response organisation, X_t includes relevant developed and current state, and B_t states containing conditions. An inherited change $\theta \to \theta'$ can change the successor relation, comparison field, physiological feasibility, or responses under the same environmental cue. Evolutionary change therefore provides a route to changed operative future geometry without an organism first selecting a different action in a hypothetical model.

The organism's own lifetime learning and hypothetical processing provide additional routes to changing determining conditions.

3.1 Differential continuation without a future model

Consider a constructed selection-only population with two faithfully inherited types and a fixed environment. Define p_i as the initial type frequencies and w_i as independently specified reproductive output per type over one generation. With positive mean output and no mutation, migration, or finite sampling, the next frequencies are

$$p_i' = \frac{p_i w_i}{\sum_j p_j w_j}.$$

For $p_1 = p_2 = 1/2$ and $w_1 = 1, w_2 = 2$, they are $1/3, 2/3$. The output definition yields the update by counting descendants and normalising; no forward comparison by either type is needed. This is a population bookkeeping instance, not an equation for an organism choosing its next action. Mutation, variable environments, transmission effects, and sampling require additional terms in a fuller model. Existing information-theoretic treatments of selection analyse these distributional changes (Frank, 2012); the connection is prior work rather than a new identification of fitness with informational or physical energy.

3.2 A measured change in reachable biological futures

In the long-term *Escherichia coli* experiment, citrate was present in the medium before a lineage evolved aerobic citrate use. Replay experiments showed that later sampled genotypes had greater propensity to evolve the trait than the ancestral and early samples, supporting historical potentiation (Blount et al., 2008). Subsequent genomic analysis identified a tandem duplication that placed a citrate transporter under an aerobically expressed promoter (Blount et al., 2012). The changed regulatory organisation enabled a physiological continuation that the earlier organisation did not implement under those conditions.

In the present framework, the resource was already part of the containing environment; changed organism structure altered its usable future geometry. The bacteria did not need to represent citrate use as a possible future and then approve it. Prior history matters through the structure it leaves and the subsequent changes that structure permits. The experimental replay and genomic reconstruction make this more specific than merely attributing an unobserved preference to the successful lineage.

Evolution is therefore the central example of changed organisation changing continuation. It also shows why hypothetical processing is one route to revision rather than its defining requirement. This is distinct from the momentum paper's maternal-inheritance grounding (Section 11), whose particular persistence claims retain their stated limits. Differential reproductive success alone does not prove that each realised action takes a strict minimum of physical energy: physiological expenditure, reproductive output, and the operative action comparison are separately grounded quantities. The minimum rule below adds that further, independently answerable hypothesis.

4 Continuation and cost geometry

Seldon's Definition 12.1 distinguishes local edge cost $c_{\mathcal{F}}(x, y)$ from accumulated cost-to-reach $C_{\mathcal{F}}(y)$, and supplies basins only after a successor policy is declared. Neither quantity is automatically a physical energy, a state potential, or Narrativium.

Definition 4.1 (Finite comparison specification)**.** *Supply a finite state space, an admissible successor set* $\mathrm{Succ}_{\mathcal{F}}(x)$*, and a feasibility predicate* $\mathrm{Feasible}_{\mathcal{F}}(x, y)$ *using declared capacity conversions. Let* $K_{\mathcal{F}}(x)$ *be the feasible successor set. Declare an effective-cost field* $q_{\mathcal{F}}(x, y)$ *with*

fixed units or comparison convention, and a deterministic resolution rule selecting one member of each nonempty tied set. Where $K_{\mathcal{F}}(x) \neq \varnothing$ and every candidate has defined finite cost, the minimum continuation is

$$b_{\mathcal{F}}(x) = \mathrm{Resolve}_{\mathcal{F}}\left(\arg\min_{y \in K_{\mathcal{F}}(x)} q_{\mathcal{F}}(x, y)\right).$$

An empty feasible set terminates this policy. A fixed point requires an explicitly supplied self-loop. If a comparison cost is undefined, this minimum rule is undefined; no missing value silently becomes zero or infinity.

One possible grounding is a state score plus a priced transition:

$$q_{\mathcal{F}}(x, y) = U_{\mathcal{F}}(y) + \lambda_{\mathcal{F}} c_{\mathcal{F}}(x, y),$$

where U, c, and the conversion λ are independently supplied and commensurable. This formula is a model specification, not an identity between preference and physical energy. Setting $U = 0$, $\lambda = 1$ gives the cheapest-successor instance of *Seldon*. A multi-dimensional account can instead supply a priority order; a scalar field then requires an explicit conversion. Physical expenditure and comparison score should be separately reported even when a model relates them.

Hypothesis 4.2 (Minimum continuation)**.** *Within a declared grounding, action-bearing continuation realises the minimum of the independently specified effective-cost field over its operative comparison domain, with the declared tie rule. The rule is instantiated in the system's dynamics; it need not be recomputed as conscious inference at each action.*

This gives the proposed "lowest-energy reachable state in the frame" statement a precise instance: the next realised state is the lowest defined effective-cost feasible successor. A history-level comparison over a longer horizon is a different instance and needs a declared history score and the mechanism carrying that comparison into action. No claim that local minima yield the globally cheapest destination follows. If q is not physical energy, a joule-denominated interpretation needs additional grounding.

The identity claim places the decision within continuation; the hypothesis restricts which continuation occurs. Assigning the observed successor cost zero and all others cost one after observation makes any deterministic trajectory cheapest and supplies no evidence for the hypothesis. A probability model permitting nonminimum actions also does not establish an exact minimum law; a proposed noisy or stochastic extension must state its different rule.

Proposition 4.3 (Local minimum need not minimise history cost)**.** *A minimum-edge successor policy need not realise a minimum-cost history over a horizon of two transitions.*

Proof. Supply exactly two histories $x \to a \to z$ and $x \to b \to w$, with edge costs $1, 10$ and $2, 0$, respectively, and sufficient capacity for both. Local minimisation chooses a at x; its history costs 11. The other history costs 2. No ties or unknown costs occur. □

Likewise, a cycle under a cheapest-successor policy cannot be strict descent of one state potential: the potential would have to be strictly smaller upon returning to the same state. Basins and escape ridges describe the supplied policy, not a universal gradient of accumulated expenditure. Unresolved dimensions stay NULL. Feasibility, membership, and cost remain typed separately.

The momentum paper's cost-form hypothesis (Hypothesis 6.1) supplies positive alteration cost with independently measured baseline expenditure. Its cheapest-reuse comparison requires equal baselines and continuing admissibility. A cheaper baseline can outweigh the alteration charge; a grammar change can remove a transition altogether. The present account inherits these conditions.

5 Prospective deliberation and autonomous processing

Definition 5.1 (Prospective deliberation)**.** *Deliberation is present processing of possible continuations not yet realised at the declared causal resolution. It may compare, simulate, inhibit, prepare, or commit, and can change operative conditions of later continuation. The processing itself is a realised transition with its own costs.*

One can deliberate while acting, or form a commitment about the next causal step. Reviewing a realised past concerns its present representation and future consequences. It does not change the realised history. No requirement that every action be preceded by explicit future modelling follows.

Deliberate cognitive deformation is a particular route to changed conditions. Where it influences an action or commitment, it supplies the causal basis for the narrower decision attribution of Definition 1.3. Evolutionary restructuring in Section 3 shows that changed continuation can also arise without that route. A change in behaviour is therefore insufficient by itself to infer a deliberative decision.

In the familiar System 1/System 2 language, *System 1 simply is*: the installed autonomous response organisation operates. There is no additional act of conscious approval required to turn that organisation into action. System 2 is embedded hypothetical processing that can reorganise conditions of subsequent action. Both belong to the present operative system.

The research connection is to autonomous defaults and working-memory-dependent hypothetical processing in Evans and Stanovich (2013). Their Type 1/Type 2 terminology avoids treating these as exactly two cognitive or neural systems. This is an interpretive mapping: Type 1 does not name every operative transition, and Type 2 does not name every revision. Autonomous adaptation can alter standing structure; hypothetical processing can fail to redirect it. Processing can overlap, and Type 2 need not always yield a better action. The cognitive distinction does not prove Hypothesis 4.2.

5.1 The woodlouse: choice carried in locomotor state

The woodlouse provides an organism-scale example of physical decision. In turn-alternation experiments with *Porcellio scaber*, preceding forced turns influenced subsequent free-turn direction and angle. Hughes's experiments supported an explanation in terms of bilaterally asymmetrical leg movements (BALM), with accumulated effects of preceding turns affecting the subsequent response (Hughes, 1985). The determining organisation therefore includes locomotor state carried into the apparent choice point, rather than just the visible branches of the runway.

In this framework, the next turn is the realised action-bearing continuation of the woodlouse's body in its containing environment. The preceding movement changes physical conditions of that continuation. An observer can describe left and right as alternatives, but that description does not require the animal to hold an explicit menu of alternatives, understand the route, or deliberately select a corrective turn. History can constrain action through the state it leaves in the organism. This is decision in the broad structural sense of Definition 1.1; it does not by itself warrant the narrower deliberative attribution.

The apparent choice point makes the provenance issue concrete. A turn by itself does not disclose whether route comparison or carried locomotor asymmetry produced it. Controlled changes to preceding movement provide evidence about the mechanism. The bodily trace is real; identifying its origin and explanatory role is a model's inference. This is the same distinction as Remark 1.4, applied to physical action.

The body's contribution is part of operative future geometry. Similar visible junctions need not supply identical operative conditions when preceding locomotion has left different bodily states. Changing those conditions changes the subsequent response without requiring a further chooser outside the response organisation. At this scale, the biological example connects

evolved organisation to physical action, while the language-model analogy below connects learned organisation to generated output.

Hughes left unresolved whether the asymmetry's control had a neural basis or another source such as muscle fatigue. Neural control is itself physical; the example does not depend on excluding it. The study supports locomotor asymmetry as a mechanism and requires no demonstrated explicit route comparison, but does not prove absence of every internal representation or measure a literal minimum-energy rule. Behavioural competence, mechanism, explicit understanding, and operative cost remain distinct questions.

5.2 Competent response does not require explicit understanding

The structural implication is that understanding is not an inherent prerequisite of autonomous action. An installed response can work without the agent holding an explicit account of why it works, what history installed it, or how its consequences arise. Evolutionary adaptation, training, and practice can leave operative structure that produces a response without retaining a recoverable explanation of that structure's origin. Competence and explicit causal understanding are therefore separately assessed.

Here explicit causal understanding means a usable model of the relevant relations and consequences that can support explanation, counterfactual comparison, and deliberate revision. It is neither guaranteed by successful output nor excluded by autonomous processing. If understanding instead names an implicit representation or task-sensitive competence, an autonomous system can possess it. The proposal is a non-requirement of explicit understanding, not a definition that strips every Type 1 process of all semantic or causal representation. Type 2 processing also need not yield accurate understanding: its represented relations can be incomplete or wrong.

A basic autoregressive language model provides an operational analogy. Its learned parameters and current context determine a next-token distribution, and a declared decoding operation produces continuation. Language-model training can yield task performance from this objective (Radford et al., 2019); generation need not be preceded by a separate explicit account of why the response is appropriate. Training changes the learned organisation, while inference uses that organisation with changing context and activations. The analogy concerns response generation by installed structure, rather than identity of human and Transformer mechanisms.

The analogy does not establish universal absence of understanding in language models. A sequence model trained on Othello moves developed a board-state representation that could be intervened on to change its output (Li et al., 2023). Learned continuation can therefore contain causally effective internal models even without an explicitly supplied world model. Nor does sampling from a language model establish the exact minimum policy of Definition 4.1.

This sharpens the decision-attribution problem. Fluent explanation, competent action, or a stated reason alone does not establish that an accurate causal model deliberately deformed operative geometry before the action. Such a model's existence, accuracy, and influence are different questions. A later explanation can itself be a generated continuation whose relation to the earlier action must be investigated. The account permits action without explicit understanding while leaving particular claims of understanding answerable to their declared criteria.

Initiating deliberation is itself an operative transition. A scheduled review, a learned alarm response, or a declared residual threshold can be part of an instance's installed rules. The account requires no further deliberator to choose whether to install that rule at each occurrence. Neither the minimum policy nor the cognitive mapping alone derives a universal trigger.

6 Narrative continuation and represented decision opening

The momentum paper defines free continuation as what persists where no rule writes against standing history-bearing structure (Definition 5.2). At the decision level, this yields a narrower

reading.

Definition 6.1 (Narrative default)**.** *If the continuation induced by $S_{\mathcal{F}}$ remains represented and admissible, retaining it requires zero* additional decision inference*. This assigns zero incremental inference charge to reuse of the installed decision in the declared grammar. Execution, maintenance, and monitoring can consume resources; delayed or failed implementation is possible.*

Definition 6.2 (Decision opening)**.** *Within the declared represented grammar, a standing commitment is reopened only by either a represented surprise $\sigma_{\mathcal{F}}$, or a current-frame meta-analysis $m_{\mathcal{F}}$:*

$$\text{Change}(\mathcal{F}) \implies \text{Surprise}_{\mathcal{F}}(\sigma_{\mathcal{F}}) \vee \text{MetaAnalyse}_{\mathcal{F}}(m_{\mathcal{F}}).$$

Meta-analysis is an additional decision-inference operation. Its cost and the conditions that trigger it vary by frame. It is not implementation cost.

Here the disjunction classifies reopening operations; it does not predict which meta-analysis trigger occurs. A predictive application must specify, for example, a scheduled review or a threshold on accumulated residual. It is not an exhaustive taxonomy of every cause of changed action. External intervention, capacity loss, or automatic adaptation can change operative continuation without represented reopening. A surprise can also activate an installed autonomous response; it is not automatically Type 2 processing.

Proposition 6.3 (Direct surprise revision is aperture-resolved)**.** *A surprise directly licenses revision only if it has a resolution x already available in the current aperture and represented landscape, and that resolution is admissible and coherent in the current frame:*

$$\text{SurpriseRevision}_{\mathcal{F}}(\sigma_{\mathcal{F}}, x) \implies x \in L_{\mathcal{F}} \text{ within } A_{\mathcal{F}} \wedge \Gamma_{\mathcal{F}}(x) \wedge \text{Cohere}_{\mathcal{F}}(\sigma_{\mathcal{F}}, x).$$

Proof. The first conjunct is required by the decision domain (Definition 2.2); an unrepresented resolution is not yet a candidate. The second is required by admissibility. The third states that a resolution cannot borrow its warrant from a successor frame not yet installed. Without all three, the mismatch can be represented as surprise but not as a directly selectable replacement. □

6.1 Expected non-occurrence is a negative observation

Definition 6.4 (Expected non-occurrence)**.** *Let X be a represented event and Y a represented opportunity condition or recurrence interval. The frame holds a negative observation $\neg X$ over Y only if all of the following hold:*

(i) $\mathcal{F}$ represents an expectation that X occurs under Y, such as a recurrent relation "X each Y";

(ii) Y is represented as having occurred and the relevant observation interval has elapsed;

(iii) the current aperture can distinguish non-occurrence from an undisclosed or unobserved occurrence; and

(iv) X is not represented in that interval.

Only then may $\neg X$ be treated as candidate surprise rather than as silence or missing information.

Proposition 6.5 (Non-occurrence constrains, but does not identify, change)**.** *If an expected non-occurrence satisfies Definition 6.4, it constrains the represented Seldon manifold by making continuations that require X under the observed Y non-coherent with the current observation. It does not identify a unique hidden change. A probabilistic expectation that allows absence is instead updated under its declared evidence model; absence alone does not logically eliminate a history that permits it.*

Proof. The represented recurrence and observed opportunity make continuations requiring X incompatible with the represented absence. Several distinct represented causes can remain compatible with that absence: a changed boundary condition, a changed intervention, a changed coupling, or a changed recurrence. If reliable non-occurrence cannot be distinguished from observation failure, the definition's aperture condition is unmet and a weaker evidence model is required. Hence the absence narrows the represented future geometry without determining one causal explanation. □

Remark 6.6 (Counting what did not happen). A repeated event stopping is therefore not a mere lack of data. Given the represented expectation and adequate aperture, it is data of negative form. It may be surprise and thereby open decision. Direct revision still requires the aperture-available, frame-coherent resolution of Proposition 6.3; otherwise the absence may motivate meta-analysis to distinguish the surviving manifold shapes.

Remark 6.7 (Unresolved surprise). An unresolved or frame-incoherent surprise may motivate meta-analysis. That operation may refine aperture, manifold resolution, active criteria, or the relation between them. It does not entail that the Narrative changes; it only pays to reopen comparison.

7 Target shapes and feasible selection

Definition 7.1 (Target shape)**.** *Where $\mathcal{F}$ represents a Seldon manifold, a target is an acceptable shape*

$$\mathcal{T}_{\mathcal{F},h,\rho} \subseteq \mathcal{S}_{h,\rho}(\mathcal{F}),$$

which may contain regions of future states, families of histories, relations, or constraints. It is a represented object in $\mathcal{F}$, not a hidden privileged destination.

This is the acceptable region of the Seldon theory (Definition 11.1) at the decision frame's horizon. T is local notation for that region, not an additional kind of object.

Proposition 7.2 (A target shape is not a point with an error bar)**.** *A point estimate $\hat{z} \pm \epsilon$ represents uncertainty about one intended location or value. A target shape represents acceptability: distinct configurations may be acceptable while nearby configurations may not be. The two remain different relations even when their represented regions have the same geometry.*

Proof. The first object is indexed by uncertainty around $\hat{z}$. The second is membership in $\mathcal{T}$. A disconnected, non-convex, or constraint-defined $\mathcal{T}$ cannot in general be recovered from a point and one scalar radius. □

Definition 7.3 (Eligible decision set)**.** *Let*

$$\mathcal{E}_{\mathcal{F}} = \{x \in L_{\mathcal{F}} \mid \Gamma_{\mathcal{F}}(x) \ \wedge \ \textit{any applicable Narrativium feasibility gate holds}\}.$$

A transition may be desirable yet absent from $\mathcal{E}_{\mathcal{F}}$ because its represented consequential deformation is not currently reachable.

Proposition 7.4 (Pareto comparison does not select)**.** *Let $v_{\mathcal{F}} : \mathcal{E}_{\mathcal{F}} \to \mathbb{R}^k$ be declared comparison dimensions and let $\mathcal{P}_{\mathcal{F}}$ be its non-dominated subset. Membership in $\mathcal{P}_{\mathcal{F}}$ does not select a unique candidate.*

Proof. Two candidates can be mutually non-dominated. Then Pareto comparison supplies no ordering between them. A priority order, scalarisation, lexicographic rule, or another stated selection relation is still required. □

After a decision opening, selection can ask whether an eligible transition reaches or remains within $\mathcal{T}_{\mathcal{F},h,\rho}$, then apply its stated selection relation among qualifying histories. Narrativium is still a capacity gate, not a value of one target point over another. Within this retention grammar, absent surprise and meta-analysis and with operative conditions unchanged, the standing commitment remains installed despite visible non-dominated alternatives. Its persistence is a declared rule; it does not establish that the installed action minimises any independently supplied field. A minimum-continuation grounding must separately relate that field to the installed rule.

8 Self-decision and operative revision

Proposition 8.1 (Successor landscapes are present candidates)**.** *A currently represented transformation τ may have a successor decision frame*

$$\mathcal{F}' = \tau(\mathcal{F}),$$

with changed aperture, target shape, Narrative, admissibility, or future manifold. The current frame evaluates τ only through what it represents and orders now.

Proof. Admissibility conditions are functions of antecedent state by the antecedent-state principle (Proposition 5.2). The newly installed frame cannot authorise the transition that first installs it (Corollary 5.3). Therefore a successor landscape is a current candidate, not a prior authority. □

Remark 8.2 (No homunculus and no future veto)*.* This is a structural constraint, not a claim that every human or artificial system actually makes self-revisions in this way. It permits direct self-change under a present frame's rules and capacity, but prevents a successor-only value, identity, or order from supplying the initial warrant.

Definition 8.3 (Operative revision)**.** *A revision changes causally relevant conditions of subsequent continuation: aperture, horizon, comparison fields, target constraints, standing structure, admissibility, capacity, coupling, or the transition machinery. A represented revision is a change in the model held by the agent. It is an operative revision of action only insofar as it changes conditions generating that action at the declared scale.*

Thinking of an alternative is a real present process. It need not install the relations required to realise the alternative. Training, reappraisal, inhibition, changed incentives, and commitment devices are candidate ways of changing operative conditions; their efficacy and costs need grounding. Revising an acceptable region, changing its weight, and changing the set of physically available actions are different transformations.

Proposition 8.4 (Changed continuation requires changed determining conditions)**.** *In a sufficient deterministic instance, two occurrences with identical operative state, boundary conditions, and transition rule have identical successors. A different successor therefore requires a difference in at least one of those determining conditions.*

Proof. The supplied deterministic function T has one value for fixed arguments. □

This formalises the structural constraint on changing a decision by changing its operative manifold. It is not independent evidence for minimisation. Identical visible action menus are weaker than identical determining conditions: different weights, alarm states, or carried commitments can produce different actions over the same menu. Conversely, an unexplained action difference must not be rescued by inventing an unrestricted hidden field after the event. It challenges the declared grounding or its state sufficiency.

Revision costs belong to the transition that revises the organisation. A later cheap continuation does not erase those costs. The present organisation must carry the trigger, resources, and admissibility of the first revision; the future self supplies no prior authorisation. This condition does not derive a uniquely correct response to transformative preference.

9 Worked finite instances

The following are constructed instances, not measurements of human energy or ethical value. They supply domains and costs before applying the rule. All tables use scalar effective-cost units; physical expenditure is separately declared where needed. Ties are resolved by the fixed label order $a < b$. Action endpoints are terminal unless a further edge is supplied.

9.1 Installed continuation, expanded aperture, and reappraisal

At action state x, let the admissible menu be $\{a, b\}$, with both actions feasible. The installed response implements Definition 4.1. Declare the score

$$q(a) = 1 + 7z, \qquad q(b) = 4,$$

where $z \in \{0, 1\}$ states whether a particular consequence is represented and coupled into the operative comparison. Both actions have physical execution expenditure of one resource unit. A review operation costs two resource units, and the initial resource budget is four. An installed alarm rule sends the system to review when a cue is registered, before the action transition. Review sets $z = 1$ and leaves two resource units; without a cue, the response uses $z = 0$. Thus neither initiating review nor redirecting action needs an unspecified chooser.

Operative comparison	$q(a)$	$q(b)$	Next action
Consequence omitted: $z = 0$	1	4	a
Consequence coupled: $z = 1$	8	4	b

Without the alarm, the installed response realises a with zero additional decision inference and positive execution expenditure. The alarm/review path realises b after paying the review charge. The reachable action menu stays the same; the field changes. The seven score units are not seven physical resource units. The review rule is supplied as part of this instance, not derived from the action scores.

This can model expanded aperture when a previously omitted dimension becomes represented and causally coupled. A separate reappraisal instance keeps that dimension represented throughout and changes its coefficient from zero to seven. It produces the same table by changing weighting rather than aperture. Both descriptions must identify the actual coupling that changes the response. Neither eliminates consequences that occurred while they were omitted.

9.2 A represented alternative without operative redirection

Extend the state with a displayed intention p. The observer knows the predeclared response rule uses z, not p. A display operation sets $p = b$, costs one resource unit, and leaves $z = 0$. The next action remains a. The display operation changed actual state and represented intention, but not the field or rule governing the action. This instance distinguishes knowing or describing an alternative from installing its operative conditions. It does not assume that intentions never have causal force.

9.3 Inadmissibility versus limited capacity

At a fresh x, declare $q(a) = 1, q(b) = 4$, and physical action expenditures $e(a) = 3, e(b) = 1$. Set feasibility to $e(y) \leq r$, where r is current resource budget. With $r = 3$, both are feasible and the policy realises a. With $r = 1$, the same admissible menu remains, but only b is feasible. Removing the edge to a by a separate rule change instead makes a inadmissible even with $r = 3$. The two deformations can yield the same action while changing different parts of operative geometry. The budget here is an explicit physical resource constraint; equating it with a Narrativium gate would require a separately grounded conversion.

9.4 Horizon and basins

The two-transition histories in Proposition 4.3 have totals 11 and 2, although their first edges cost 1 and 2. A supplied two-step planning operation can compute both totals and install the history through b. Give that operation a separate expenditure of one unit in the same accounting currency, with additive path and planning expenditure declared for this instance; its planned course then costs 3. Both endpoints are acceptable and both courses are feasible under a budget of twelve. This construction shows what a longer-horizon rule can implement, but does not derive its initiation or price from the local rule.

For a basin example, separately supply edges $u \to v$, $v \to u$, and $v \to w$ with local costs $1, 1, 3$, and a declared absorbing self-loop at w of cost zero. Under the local policy, $\{u, v\}$ is an invariant cycle with basin $\{u, v\}$. Its minimum escape-edge cost is three, but that edge is never taken by the unchanged cheapest-successor policy. An external intervention, constraint revision, or changed comparison field is needed to produce escape. The cycle is an admissible instance, not descent to a global minimum of a single state potential.

10 Decision quality, implementation, and outcome

The account is not an ideal-decision theory. In the retention grammar, an unchanged operative frame carries its installed commitment without reopening. A frame-valid but poor represented selection can cohere within a limited or miscalibrated aperture. An agent may pay for meta-analysis and still fail to realise the intended action. Implementation may fail while the represented selection or commitment remains unchanged. An intended action may also be realised yet lead to undesirable consequences. These are distinct diagnosis classes; an unchanged commitment does not make a failed intended action into the same realised decision.

This distinction keeps outcome learning correctly typed. A later consequence may become represented surprise, which can then have an aperture-available, frame-coherent resolution or motivate meta-analysis. It does not make the external consequence a retroactive input to the original decision.

Whole-system consequences are also distinct from the operative comparison that generated action. The informational accounting companion (Section 6) separates an actual consequence account from the model and relative weights through which an agent evaluates it. An omitted casualty, future loss, or burden on another system can occur regardless of the actor's representation. Changes to the actor's own organisation, including moral injury and subsequent self-modelling, also belong to the consequence account. They can alter later operative fields without making the original internal score identical to whole-system cost. Action can be a minimum under a grounded operative field and still cause large externally measurable losses.

11 Operational commitments and research neighbours

The operational commitment is comparative. For a stated population, frame class, target shape, horizon, baseline, and inference budget, the grammar must be tested for:

(1) predictive value for persistence, surprise, meta-analysis, revision, and decision-relevant consequences;

(2) decision value for target-shape membership or other declared objectives;

(3) cost of representation, review, and meta-analysis; and

(4) transfer limits across aperture, coupling, and horizon.

A failure to improve these relative to a baseline limits operational usefulness in that scope. A repeated nonminimum response under independently grounded sufficient state, fields, constraints, and rules challenges the minimum-continuation hypothesis for that grounding. The identity claim alone does not distinguish such responses and must not substitute for this test.

11.1 A discriminating proposed study

Hold the available action menu fixed and specify an alarm/review operation that changes an independently measured consequence representation or its operative weighting. Fit a declared field, trigger rule, revision expenditure, and state description on a calibration set, then freeze them. On held-out cases, predict persistence without effective revision and a switch where revision reverses the feasible ordering. Compare the prediction with a fixed-frame model that has no revision term, accounting for the extra parameters and inference expenditure.

The proposal needs a manipulation check that measures the specified operative change independently of the target action; a verbal statement of intention alone is insufficient. If the field crosses the preregistered boundary but the response repeatedly fails to switch under the declared conditions, that counts against the grounding. A probabilistic observation model can specify measurement uncertainty in advance; it must not assign arbitrary fields to each failure. A demonstration only that attention changes behaviour would not uniquely support this theory. Exact recovery of a brain mechanism is not required to propose the principle, but evidence must constrain possible groundings and distinguish the claimed rule from competing accounts.

11.2 Research neighbours and contribution

Dual-process cognition supplies the distinction between autonomous responses and hypothetical processing discussed in Section 5 (Evans and Stanovich, 2013). The present account interprets both as embedded continuations and specifies how hypothetical processing can alter operative future geometry. That connection does not establish a universal brain architecture or minimum-cost dynamics.

Several other literatures are close but non-equivalent. Resource-rational metareasoning treats additional reasoning as costly and models selection of reasoning processes (Godara et al., 2024); it does not define Narrative or this opening condition. Default research establishes heterogeneous effects of preselected options (Jachimowicz et al., 2019); it does not establish a general history-bearing Narrative law. Active inference supplies a Bayesian policy model using prediction and preference models (da Costa et al., 2024); GE residual is not thereby variational free energy, prediction error, or Bayesian surprise. Multiobjective optimisation confirms that Pareto-optimal alternatives require additional preference or selection information for final choice (Pajasmaa et al., 2025). AGM belief revision provides a formal neighbour for retention and change of commitments (Alchourrón et al., 1985); it is not identical to Narrative revision.

The contribution is the integration of decision as embedded continuation, represented futures, and paid restructuring within the series' framework. The finite policy and examples expose the supplied premises. They claim no priority over planning, metareasoning, dynamical accounts of cognition, or dual-process theory, and derive no general neural or substrate-independent energy law from those neighbouring results.

12 Limits

The paper proposes an ontological interpretation and supplies implementable finite dynamics under declared premises. It does not establish that a nervous system, organisation, country, or artificial agent instantiates the complete grammar or the exact minimum rule. The interpretation requires neither conscious choice nor a complete self-model; its explanatory value depends on grounding more restrictive dynamics than the identity alone supplies.

No universal scalar energy, global optimisation theorem, interpersonal Narrativium accounting, target shape, meta-analysis trigger, or neural mechanism is derived. The local construction needs known comparison costs, nonempty feasible domains, and declared tie rules. Partial observations cannot warrant complete-state predictions. A represented future, another agent's model, and actual operative organisation remain distinct. At collective scale the account uses declared coupling among partial frames, rather than a categorical group mind.

The central proposal is therefore precise but conditional: realised decision is action-bearing continuation; deliberation is present processing of unrealised futures; effective redirection changes operative conditions. Minimum continuation adds an independently answerable hypothesis. Its human application remains open to bounded comparative investigation.

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